

Juan Manuel Young Hoyos

SOFTWARE ENGINEER, FULL-STACK | TYPESCRIPT & RUST | ROBOTICS, REAL-TIME & WEBASSEMBLY

Medellín, Colombia · Open to relocating to Amsterdam (on-site) · English B2+

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Summary

Full-stack generalist who ships production systems in **TypeScript**, Go, Python and **C++** (including refactoring a **C++ WebAssembly** codebase) and writes **Rust** for robotics — **lidar SLAM**, sensor drivers and embedded cross-compilation — happiest on problems that cut across the stack. I build **real-time, telemetry-driven** systems over **MQTT**, Kafka and message queues, design the architecture and the abstractions rather than just the features, and debug into unfamiliar layers to find the actual cause. Strong CS and **geometry/linear-algebra** foundation. I run **LLM agents in parallel** for building, review and investigation, and have led delivery in chaotic environments against hard deadlines.

Skills

Languages	TypeScript, Rust, Python, Go, C++, JavaScript, Nim · WebAssembly (legacy C++ WASM refactor)
Systems & Real-Time	Telemetry (Rerun), serial hardware, aarch64 cross-compile, MQTT, WebSockets, TCP/IP, Kafka, Linux, Docker
Frontend	React, React Native, Next.js · monitoring & control dashboards, state management, frontend performance
Geometry & Simulation	SLAM, pose estimation, occupancy grids, geohash algorithms, 3D transforms, Godot, device simulation
AI Agents	Claude Code: building, code review, security checks, incident reports — validated with engineering judgement

Open-Source Projects

Project Olivaw · github.com/Project-Olivaw

Open source

CREATOR — ROBOTICS IN RUST

2025 – Present

- Wrote **olivaw-slam**, 2D lidar **SLAM** in pure **Rust** — correlative scan matching, keyframing, a log-odds **occupancy grid** and **pose estimation** (2D transforms and linear algebra) — deliberately with no ROS2 or C++ dependencies.
- Wrote **olivaw-lidar**, a pure-Rust driver talking to **RPLIDAR hardware over serial** (no C++ SDK or FFI), cross-compiled to **aarch64** boards, streaming live scans into **Rerun** for **real-time telemetry**; plus **olivaw-cli**, a developer tool vendoring editable Rust components.

Experience

Okorum Technologies

Medellín, Colombia (Hybrid)

TECHNICAL LEAD

Dec. 2025 – Present

- Chose the core architecture and abstractions for a national-scale platform, shipped **ahead of a hard government deadline** — a distributed **TypeScript** monorepo coordinating over **message queues**, **React** front end in the same repo.
- Made an unreliable external dependency survivable with **fault tolerance and resumability** in the queue system, plus **Redis** caching and **telemetry** (logs, traces, metrics) to see what every stage is doing.
- Owned **distributed data storage and sync** across services (MongoDB, Redis) and drove problems start-to-finish with a cross-functional team, without external project management.

Publicis Global Delivery (PGD) — Digitas / Razorfish

Colombia (Remote — New York)

FRONTEND / REACT NATIVE DEVELOPER

Dec. 2023 – Nov. 2025

- Shipped **production** software at scale for US retailers — **React Native** for Wegmans (+30% engagement) and a **Next.js** redesign — plus **Unicorn AI**, an internal platform spanning its **React** front end and **Python** backend.
- Cut critical bugs **25%** (Jest, **Dynatrace** runtime monitoring) and release-cycle time **50%** via **CI/CD** for a **React Native** E-commerce retail in the US.

Grisú · ArcDesign

Colombia / Spain (Remote)

FOUNDER / FULL STACK & GAME DEVELOPER (SELF-EMPLOYED)

Jul. 2023 – Present

- Designed a spatial algorithm using **hashmaps and geohashes** and drove the architecture around **MQTT over TCP/IP**, running device **simulations in Python** before integrating the logic into a client for **near real-time** behaviour.
- Ran my own practice end to end — architecture consultancy for client organisations and a **Godot** game (3D transforms and geometry) — and compressed a 4-month plan into weeks on **EcoDrive (Kafka, Python)**.

Design Systems Inno · Aligo · Geta Club Play

Medellín, Colombia

PLATFORM & AUTOMATION — EARLIER, SYSTEMS, SECURITY & QA

Feb. 2021 – Jun. 2023

- Architected a decoupled, asynchronous system on **RabbitMQ** where clients track long-running jobs by ticket ID instead of blocking, with internal APIs in **Go** and **Python** on MongoDB, and internal tools pushing live status over **WebSockets**.
- Refactored a legacy **C++ WebAssembly** application (+45% code quality), wrote low-level tooling in **Go, Python** and Nim, and **reverse-engineered** Android binaries and network systems to find root causes (vulnerabilities **-50%** in 4 months).
- Chased failures that only appear **under sustained load** with **stress and long-running tests** across 7 products, capturing 100+ defects, while leading a 4-person team on **Linux**.

Education

EAFIT University

Medellín, Colombia

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Jan. 2018 – Dec. 2025

- Full scholarship “Ser Pilo Paga” (national merit-based award).